(11) $x=\frac{1-t^2}{1+t^2}, y=\frac{2t}{1+t^2}$ then $\frac{dy}{dx}$ is

(1) $-\frac{y}{x}$ (2) $\frac{y}{x}$ (3) $-\frac{x}{y}$ (4) $\frac{x}{y}$

(12) If $x=a\sin\theta$ and $y=b\cos\theta$, then $\frac{d^2y}{dx^2}$ is

(1) $\frac{a}{b^2}\sec^2\theta$ (2) $-\frac{b}{a}\sec^2\theta$ (3) $-\frac{b}{a^2}\sec^3\theta$ (4) $-\frac{b^2}{a^2}\sec^3\theta$

(13) The differential coefficient of $\log_{10} x$ with respect to $\log_x 10$ is

(1) 1 (2) $-(\log_{10} x)^2$ (3) $(\log_x 10)^2$ (4) $\frac{x^2}{100}$

(14) If $f(x)=x+2$, then $f'(f(x))$ at $x=4$ is

(1) 8 (2) 1 (3) 4 (4) 5

(15) If $y=\frac{(1-x)^2}{x^2}$, then $\frac{dy}{dx}$ is

(1) $\frac{2}{x^2}+\frac{2}{x^3}$ (2) $-\frac{2}{x^2}+\frac{2}{x^3}$ (3) $-\frac{2}{x^2}-\frac{2}{x^3}$ (4) $-\frac{2}{x^3}+\frac{2}{x^2}$

(16) If $pv=81$, then $\frac{dp}{dv}$ at $v=9$ is

(1) 1 (2) – 1 (3) 2 (4) –2

(17) If $f(x)=\begin{cases}x-5 & \text{if } x\le 1\\ 4x^2-9 & \text{if } 1<x<2\\ 3x+4 & \text{if } x\ge 2\end{cases}$, then the right hand derivative of $f(x)$ at $x=2$ is

(1) 0 (2) 2 (3) 3 (4) 4

(18) It is given that $f'(a)$ exists, then $\lim_{x\to a}\frac{xf(a)-af(x)}{x-a}$ is

(1) $f(a)-af'(a)$ (2) $f'(a)$ (3) $-f'(a)$ (4) $f(a)+af'(a)$

(19) If $f(x)=\begin{cases}x+1, & \text{when } x<2\\ 2x-1 & \text{when } x\ge 2\end{cases}$, then $f'(2)$ is

(1) 0 (2) 1 (3) 2 (4) does not exist

(20) If $g(x)=(x^2+2x+1)f(x)$ and $f(0)=5$ and $\lim_{x\to 0}\frac{f(x)-5}{x}=4$, then $g'(0)$ is

(1) 20 (2) 14 (3) 18 (4) 12

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